

Learning Hazing to Dehazing: Towards Realistic Haze Generation for Real-World Image Dehazing

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Abstract

Existing real-world image dehazing methods primarily attempt to fine-tune pre-trained models or adapt their inference procedures, thus heavily relying on the pre-trained models and associated training data. Moreover, restoring heavily distorted information under dense haze requires generative diffusion models, whose potential in dehazing remains underutilized partly due to their lengthy sampling processes. To address these limitations, we introduce a novel hazing-dehazing pipeline consisting of a Realistic Hazy Image Generation framework (HazeGen) and a Diffusion-based Dehazing framework (DiffDehaze). Specifically, HazeGen harnesses robust generative diffusion priors of real-world hazy images embedded in a pre-trained text-to-image diffusion model. By employing specialized hybrid training and blended sampling strategies, HazeGen produces realistic and diverse hazy images as high-quality training data for DiffDehaze. To alleviate the inefficiency and fidelity concerns associated with diffusion-based methods, DiffDehaze adopts an Accelerated Fidelity-Preserving Sampling process (AccSamp). The core of AccSamp is the Tiled Statistical Alignment Operation (AlignOp), which can provide a clean and faithful dehazing estimate within a small fraction of sampling steps to reduce complexity and enable effective fidelity guidance. Extensive experiments demonstrate the superior dehazing performance and visual quality of our approach over existing methods. The code is available at <https://github.com/ruiyi-w/Learning-Hazing-to-Dehazing>.

1. Introduction

Images captured under hazy conditions frequently exhibit color distortion and detail loss, significantly impairing the performance of downstream vision tasks. The formation of hazy images is typically modeled by the physical scatter-

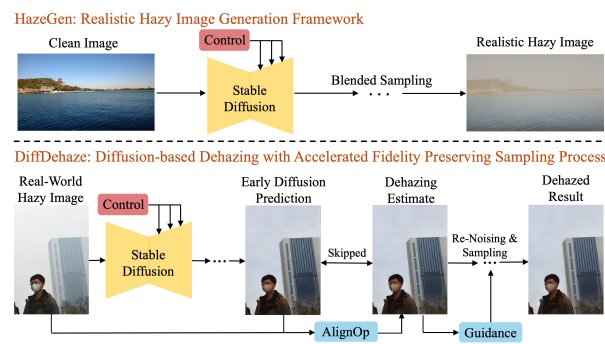


Figure 1. Overview of the proposed pipeline. HazeGen utilizes a pre-trained text-to-image diffusion model to generate realistic hazy images, which serve as the training data for DiffDehaze. DiffDehaze adopts an Accelerated Fidelity-Preserving Sampling process (AccSamp) that effectively reduces sampling steps while producing superior dehazing results with enhanced fidelity.

ing model [30, 32]:

$$I(x) = J(x) \cdot t(x) + A \cdot (1 - t(x)), \quad (1)$$

where $I(x)$ represents a hazy image and $J(x)$ is its clean counterpart. The parameters A and $t(x)$ denote the global atmospheric light and the transmission map, respectively.

Estimating the parameters, particularly the transmission map, from a single hazy image is inherently ill-posed. Early methods incorporated statistical priors of clean images, such as the Dark Channel Prior (DCP) [16] and Non-Local Prior (NLP) [2], to constrain solutions. However, these priors often fail to generalize across diverse real-world scenarios, resulting in unsatisfactory dehazing outcomes and noticeable artifacts. With the rise of Convolutional Neural Networks (CNNs), numerous learning-based approaches emerged [3, 9, 15, 27, 28, 38, 50]. Trained on large-scale datasets, these methods have achieved remarkable advancements. However, collecting large-scale, perfectly aligned real-world hazy and clean image pairs is nearly impossible. Although some real-world datasets have been constructed [1, 4], their scale and diversity remain insufficient

从加雾学习去雾：面向真实世界图像去雾的真实雾气生成

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摘要

现有的真实世界图像去雾方法主要尝试微调预训练模型或调整其推理流程，因此严重依赖预训练模型及其相关训练数据。此外，在浓雾下恢复严重失真的信息需要生成式扩散模型，而扩散模型在去雾中的潜力仍未得到充分利用，部分原因在于其采样过程冗长。为了解决这些局限性，我们提出了一种新颖的加雾-去雾流水线，包括一个真实有雾图像生成框架（HazeGen）和一个基于扩散的去雾框架（DiffDehaze）。具体来说，HazeGen 利用嵌入在预训练文本到图像扩散模型中的真实世界有雾图像的鲁棒生成式扩散先验。通过采用专门的混合训练和混合采样策略，HazeGen 生成真实且多样的有雾图像，作为 DiffDehaze 的高质量训练数据。为了缓解基于扩散的方法所存在的低效率和保真度问题，DiffDehaze 采用了加速保真采样过程（AccSamp）。AccSamp 的核心是分块统计对齐操作（AlignOp），它能在极少的采样步数内提供干净且忠实的去雾估计，从而降低复杂度并实现有效的保真度引导。大量实验表明，我们的方法在去雾性能和视觉质量上均优于现有方法。代码可在

<https://github.com/ruiyi-w/Learning-Hazing-to-Dehazing> 获取。

1. 引言

在有雾条件下拍摄的图像经常表现出颜色失真和细节丢失，显著损害了下游视觉任务的性能。有雾图像的形成通常由物理散射模型描述，

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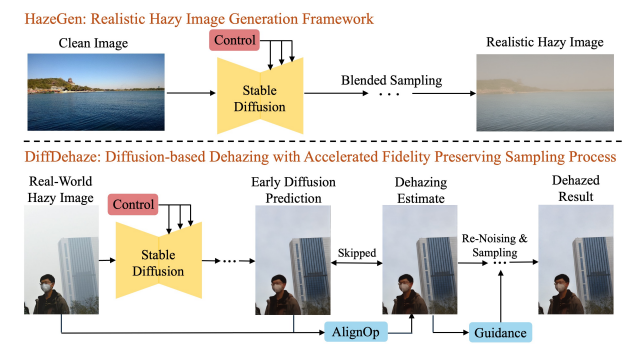


图1. 所提流程概览。HazeGen利用预训练的文本到图像扩散模型生成真实感有雾图像，作为DiffDehaze的训练数据。DiffDehaze采用加速保真采样

过程（AccSamp），有效减少采样步数，同时产出具有增强保真度的优越去雾结果。

散射模型 [30, 32]:

$$I(x) = J(x) \cdot t(x) + A \cdot (1 - t(x)), \quad (1)$$

其中 $I(x)$ 表示有雾图像， $J(x)$ 为其对应的清晰图像。参数 A 和 $t(x)$ 分别表示全局大气光和透射率图。

从单张有雾图像中估计参数，特别是透射率图，本质上是病态问题。早期方法引入了干净图像的统计先验，例如暗通道先验（DCP）[16]和非局部先验（NLP）[2]，来约束解。然而，这些先验往往难以泛化到多样化的真实世界场景中，导致去雾效果不理想并出现明显伪影。随着卷积神经网络（CNNs）的兴起，大量基于学习的方法应运而生 [3, 9, 15, 27, 28, 38, 50]。这些方法在大规模数据集上训练，取得了显著进展。然而，收集大规模、完美对齐的真实世界有雾与清晰图像对几乎是不可能的。尽管已经构建了一些真实世界数据集 [1, 4]，但其规模和多样性仍然不足

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for training robust deep neural networks. Consequently, most current models rely on synthetic datasets, leading to performance degradation when the models are applied to real-world hazy scenes.

Recognizing this limitation, recent studies have increasingly focused on real-world image dehazing. Several studies [6, 44] reintroduced prior knowledge to fine-tune pre-trained models, while others [7, 47] adapted their inference procedures. Although these methods led to performance gains, they remain significantly dependent on pre-trained models and, consequently, the quality of pre-training data. This dependency highlights the necessity of developing large-scale realistic hazy image generation approaches, which are currently lacking. Moreover, heavily hazed images involve significant information loss, posing challenges to conventional *enhancement*-based methods that lack generative flexibility to recover such information [47]. While diffusion models have demonstrated remarkable success in image generation [34, 37, 39], their application to image dehazing remains limited partly due to lengthy sampling processes, underscoring the importance of efficient sampling strategies.

To address these challenges, we propose a hazing-dehazing pipeline comprising a Realistic Hazy Image Generation framework (HazeGen) and a Diffusion-based Dehazing framework (DiffDehaze). Recent advancements in text-to-image diffusion models [34, 37, 39], trained on extensive datasets containing diverse hazy scenarios, have demonstrated remarkable capabilities in producing high-quality images. By providing suitable prompts, these pre-trained diffusion models can naturally serve as effective generators of realistic hazy images—an insight central to our approach. To enable conditional generation, HazeGen utilizes IRControlNet [26], which injects conditional information into the fixed diffusion prior. To further enhance the realism of generated hazy images, HazeGen adopts hybrid conditional and unconditional training objectives and a blended sampling strategy. During sampling, incorporating a small fraction of unconditional predictions can significantly enhance the realism and variety of generated images.

DiffDehaze is trained using high-quality data produced by HazeGen and employs an Accelerated Fidelity-Preserving Sampling process (AccSamp). The core of AccSamp is the Tiled Statistical Alignment Operation (AlignOp), whose effect is shown in Figure 4. Drawing inspiration from adaptive instance normalization [19] and image-level normalization [7], AlignOp substitutes the mean and variance of the hazy image with that of an early diffusion prediction in patches to produce a rough dehazing estimate. Although the early diffusion prediction is blurry, AlignOp can transform it into an effective and faithful dehazing estimate, which allows skipping intermediate sampling steps and advancing directly to a later step. To further

enhance fidelity—especially in lightly hazy regions—we additionally equip AccSamp with a haze density-aware fidelity guidance mechanism.

Our contributions are summarized as follows:

- ◊ We introduce a novel Realistic Hazy Image Generation framework (HazeGen) that leverages generative diffusion priors to produce highly realistic hazy training data, significantly boosting the performance of DiffDehaze.

- ◊ We propose an Accelerated Fidelity-Preserving Sampling process (AccSamp) for DiffDehaze, which reduces the sampling steps and enhances fidelity of recovered images.

- ◊ Extensive quantitative and qualitative experiments validate that our proposed pipeline achieves superior performance compared to state-of-the-art methods.

2. Related Works

2.1. Single Image Dehazing

Early approaches to single-image dehazing [2, 12, 13, 16, 42, 55] typically reconstruct haze-free images by estimating the transmission map in the physical scattering model with statistical priors of clean images. For example, He *et al.* [16] introduced the influential Dark Channel Prior (DCP), based on the observation that the dark channel of a clean image typically approaches zero intensity. Other effective priors include the Non-Local Prior (NLP) [2] and the Color Attenuation Prior (CAP) [55]. Despite their effectiveness, these handcrafted priors generally struggle to cover the complexity and diversity of real-world imagery, for instance, DCP fails for prominent white objects, resulting in degraded dehazing effect and visible artifacts.

With the emergence of Convolutional Neural Networks (CNNs) and the availability of large-scale synthetic datasets, learning-based dehazing methods [9, 10, 15, 27, 28, 35, 36, 52] have become increasingly popular, paralleling advances in other low-level vision tasks [11, 14, 20, 24, 25, 53, 54]. Particularly, Liu *et al.* [27] proposed an attention-based multi-scale gridnetwork, and Qiu *et al.* [36] developed a Transformer variant with linear computational complexity to efficiently harness Transformer architectures for image dehazing. However, although learning-based techniques achieve superior performance on synthetic datasets, their reliance on synthetic training data and the substantial domain gap between synthetic and real-world hazy images cause significant performance degradation on real-world hazy images.

2.2. Real-World Image Dehazing

Due to the challenges and practical significance of real-world image dehazing, recent research has increasingly emphasized methods tailored for real-world scenarios. Early studies [41, 49] explored CycleGAN-based frameworks.

以训练鲁棒的深度神经网络。因此，当前大多数模型依赖合成数据集，导致模型应用于真实世界有雾场景时性能下降。

认识到这一局限，近年来的研究越来越关注真实世界图像去雾。一些研究 [6, 44]重新引入先验知识来微调预训练模型，而另一些研究则 [7, 47] 调整其推理过程。尽管这些方法带来了性能提升，但它们仍然高度依赖预训练模型，进而依赖预训练数据的质量。这种依赖凸显了开发大规模真实感有雾图像生成方法的必要性，而目前这类方法还很缺乏。此外，重度有雾图像涉及严重的信息丢失，给缺乏生成灵活性以恢复此类信息的传统增强类方法带来了挑战 [47]。虽然扩散模型已在图像生成中展现出显著的成功 [34, 37, 39]，但其在图像去雾中的应用仍然有限，部分原因在于冗长的采样过程，这凸显了高效采样策略的重要性。

为解决这些挑战，我们提出了一种加雾-去雾流程，包括一个真实感有雾图像生成框架（HazeGen）和一个基于扩散的去雾框架（DiffDehaze）。文本到图像扩散模型 [34, 37, 39]，近期在大规模包含多样化有雾场景的数据集上完成训练，已展现出生成高质量图像的卓越能力。通过提供合适的提示词，这些预训练扩散模型可以自然地充当真实感有雾图像的有效生成器——这正是我们方法的核心思路。为实现条件生成，HazeGen利用IRControlNet [26]，将条件信息注入固定的扩散先验中。为进一步提升生成有雾图像的真实感，HazeGen采用条件与无条件训练目标相结合的混合策略以及混合采样策略。在采样过程中，融入少量无条件预测可以显著提高生成图像的真实感和多样性。

DiffDehaze使用HazeGen生成的高质量数据进行训练，并采用加速保真采样过程（AccSamp）。AccSamp的核心是分块统计对齐操作（AlignOp），其效果如图4所示。受自适应实例归一化 [19]和图像级归一化 [7]，的启发，AlignOp用早期扩散预测的均值和方差按图像块替换有雾图像的均值和方差，从而生成粗略的去雾估计。尽管早期扩散预测是模糊的，AlignOp仍能将其转化为有效且保真的去雾估计，从而允许跳过中间采样步骤，直接推进到更靠后的步骤。为了进一步

增强保真度——尤其是在轻度有雾区域——我们还为AccSamp配备了一种雾密度感知的保真引导机制。

我们的贡献总结如下：

- ◊ 我们提出了一种新颖的真实有雾图像生成框架（HazeGen），该框架利用生成扩散先验来生成高度逼真的有雾训练数据，显著提升了DiffDehaze的性能。

- ◊ 我们为DiffDehaze提出了一种加速保真采样过程（AccSamp），该过程减少了采样步数并提升了恢复图像的保真度。

- ◊ 大量定量和定性实验验证了我们所提出的流程相较于最先进方法取得了更优越的性能。

2. 相关工作

2.1. 单幅图像去雾

早期的单图像去雾方法 [2, 12, 13,16, 42, 55]通常通过利用清晰图像的统计先验来估计物理散射模型中的透射率图，从而重建无雾图像。例如，He等人 [16]提出了具有影响力的暗通道先验（DCP），其基于这样一个观察：清晰图像的暗通道通常趋近于零强度。其他有效的先验还包括非局部先验（NLP） [2] 和颜色衰减先验（CAP） [55]。尽管这些手工设计的先验具有一定效果，但它们通常难以覆盖真实世界图像的复杂性和多样性，例如，DCP在显著白色物体上会失效，从而导致去雾效果下降并出现可见伪影。

随着卷积神经网络（CNNs）的出现以及大规模合成数据集的可用性，基于学习的去雾方法 [9, 10, 15, 27,28, 35, 36, 52] 变得越来越流行，与其他低层视觉任务 [11, 14, 20, 24, 25, 53, 54]的发展趋势相呼应。特别是，Liu等人 [27] 提出了一种基于注意力的多尺度网格网络，而Qiu等人 [36] 开发了一种具有线性计算复杂度的Transformer变体，以高效地将Transformer架构用于图像去雾。然而，尽管基于学习的技术在合成数据集上取得了优越的性能，但它们对合成训练数据的依赖以及合成有雾图像与真实世界有雾图像之间存在显著的域差距，导致其在真实世界有雾图像上的性能明显下降。

2.2. 真实世界图像去雾

由于真实世界图像去雾的挑战性和实际意义，近年来的研究越来越强调针对真实世界场景定制的方法。早期研究 [41, 49]探索了基于CycleGAN的框架。

Shao *et al.* [41], for example, introduced a CycleGAN-based domain adaptation approach to map images from the synthetic domain to the real domain. Nevertheless, these frameworks usually have complex and unstable training processes, as well as mode collapse issues. Another popular research direction [6, 23] is to integrate prior knowledge into the fine-tuning process of pre-trained models. For instance, Chen *et al.* [6] proposed PSD, a framework converting pre-trained dehazing networks into physically informed ones and fine-tuning them via three statistical priors. Other studies [7, 47] target improvements during inference. For example, Chen *et al.* [7] proposed a feature adaptation module designed to recalibrate encoder features during inference. Despite the advances, these techniques remain inherently dependent on the quality of the pre-training data, underscoring the necessity of robust methods capable of generating realistic and diverse hazy images. Prior attempts [22, 47] have utilized the physical scattering model and depth maps for haze synthesis. However, the physical scattering model can’t adequately represent the intricate real-world haze formation process, resulting in unrealistic and homogeneous hazy images. In contrast to explicit physical modeling, our work explores robust generative priors embedded in a pre-trained text-to-image diffusion model.

3. Methodology

The central idea of this work is to leverage the generative diffusion prior of natural hazy images. HazeGen and DiffDehaze are built upon the architectural framework introduced by DiffBIR [26], employing a fixed Stable Diffusion model and a IRControlNet for conditional information injection. IRControlNet is initialized from the UNet’s encoder and modulates features in its decoder. HazeGen further applies specialized hybrid training and blended sampling strategies, whereas DiffDehaze adopts the AccSamp sampling process for enhanced efficiency and fidelity. In the following sections, we briefly review diffusion models and then present the detailed methodologies of HazeGen and DiffDehaze.

3.1. Preliminary: Diffusion Models

Since diffusion models form the foundation for HazeGen and DiffDehaze, we briefly review key concepts of conditional denoising diffusion models [18, 40]. HazeGen aims to model the conditional distribution $P(x|y)$, where x and y denote corresponding hazy and clean images, while DiffDehaze inversely models $P(y|x)$. Diffusion models have two fundamental processes: the forward diffusion process and the reverse denoising process.

The Forward Process. The forward process progressively adds Gaussian noise to an encoded target image $z_0 = z$ according to a predefined variance schedule β_t . This process

can be succinctly formulated as:

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad (2)$$

where z_t is the noised image, $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \sum_{s=1}^t \alpha_s$ and $\epsilon \sim \mathcal{N}(0, 1)$.

The Reverse Process. The reverse process aims to reconstruct the original image by denoising. Specifically, given encoded condition c , a model ϵ_θ learns to predict the added noise ϵ at each timestep t . At a randomly sampled timestep t , noise ϵ is introduced to produce z_t . The model is trained to minimize the simplified denoising objective [18]:

$$\mathcal{L} = \mathbb{E}_{z,c,t,\epsilon \sim \mathcal{N}(0,1)} [\|\epsilon - \epsilon_\theta(z_t, c, t)\|_2^2]. \quad (3)$$

To leverage rich generative priors, both HazeGen and DiffDehaze are built upon the pre-trained Stable Diffusion model [39]. Unlike traditional diffusion methods operating in RGB pixel space, Stable Diffusion performs the diffusion and denoising processes in a low-dimensional latent space created by a separate autoencoder.

3.2. Realistic Hazy Image Generation Framework

Though equipped with powerful generative priors, HazeGen must effectively incorporate conditional information from clean images. A straightforward solution would be training HazeGen on paired synthetic images. However, we observe that this method rapidly degenerates the generative priors, as the simple haze synthesis with the physical scattering model is easily recognizable and thus is learned by the model. As a result, generated hazy images closely resemble synthetic data, causing a significant domain gap with real-world images. To alleviate this issue, we propose specialized hybrid training and blended sampling algorithms.

Hybrid Training. To preserve the ability to generate realistic hazy images while gradually enhancing content consistency, we introduce the hybrid training objective. Specifically, the conditional generation objective uses synthetic image pairs, guiding the model to build content relationships between generated hazy images and corresponding clean images. Conversely, the unconditional objective employs unlabeled real-world hazy images, helping HazeGen maintain and further enhance its capability for realistic haze generation, thus preventing catastrophic forgetting. Combining the two objectives, the hybrid training loss is:

$$\mathcal{L} = p \mathbb{E}_{z^s, c^s, t, \epsilon} [\|\epsilon - \epsilon_\theta(z_t^s, c^s, t)\|_2^2] + (1 - p) \mathbb{E}_{z^r, t, \epsilon} [\|\epsilon - \epsilon_\theta(z_t^r, \emptyset, t)\|_2^2], \quad (4)$$

where encoded synthetic image pairs are represented by (z^s, c^s) , while encoded real-world hazy images are denoted as z^r . p is a tradeoff parameter determining the probability to apply the conditional objective.

Shao 等人 [41], 例如, 引入了一种基于CycleGAN的领域自适应方法, 将图像从合成域映射到真实域。然而, 这些框架通常具有复杂且不稳定的训练过程, 以及模式崩溃问题。另一个流行的研究方向 [6, 23] 是将先验知识融入预训练模型的微调过程中。例如, Chen 等人 [6] 提出了PSD, 这一框架将预训练去雾网络转换为具有物理信息的网络, 并通过三个统计先验进行微调。其他研究 [7, 47]则着眼于推理阶段的改进。例如, Chen 等人 [7]提出了一种特征自适应模块, 旨在推理过程中重新校准编码器特征。尽管取得了这些进展, 但这些技术本质上仍依赖于预训练数据的质量, 这凸显了能够生成真实且多样的有雾图像的稳健方法的必要性。此前的尝试 [22, 47] 利用物理散射模型和深度图进行雾合成。然而, 物理散射模型无法充分表征复杂的真实世界雾形成过程, 导致生成的有雾图像不真实且同质化。与显式物理建模相比, 我们的工作探索了嵌入在预训练文本到图像扩散模型中的稳健生成先验。

3. 方法论

本工作的核心思想是利用自然有雾图像的生成式扩散先验。HazeGen和DiffDehaze基于DiffBIR引入的架构框架构建 [26], , 采用固定的Stable Diffusion模型和IRControlNet进行条件信息注入。IRControlNet从UNet的编码器初始化, 并在其解码器中调制特征。HazeGen进一步应用专门的混合训练和混合采样策略, 而DiffDehaze采用AccSamp采样过程以提升效率和保真度。在以下各节中, 我们简要回顾扩散模型, 然后详细介绍HazeGen和DiffDehaze的方法论。

3.1. 预备知识：扩散模型

由于扩散模型是HazeGen和DiffDehaze的基础, 我们简要回顾条件去噪扩散模型的关键概念 [18, 40]。HazeGen旨在对条件分布 $P(x|y)$ 进行建模, 其中 x 和 y 分别表示对应的有雾图像和清晰图像, 而 DiffDehaze则反向建模 $P(y|x)$ 。扩散模型有两个基本过程: 前向扩散过程和反向去噪过程。

前向过程。前向过程根据预定义的方差调度 β_t , 逐步向编码目标图像 $z_0 = z$ 添加高斯噪声。这一过程

可以简洁地表示为:

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad (2)$$

其中 z_t 是加噪图像, $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \sum_{s=1}^t \alpha_s$ 和 $\epsilon \sim \mathcal{N}(0, 1)$ 。**逆向过程。**逆向过程旨在通过去噪来重建原始图像。具体而言, 给定编码条件 c , 模型 ϵ_θ 学习在每个时间步 t 预测所添加的噪声 ϵ 。在随机采样的时间步 t , 引入噪声 ϵ 以生成 z_t 。该模型通过最小化简化的去噪目标 [18]进行训练:

$$\mathcal{L} = \mathbb{E}_{z,c,t,\epsilon \sim \mathcal{N}(0,1)} [\|\epsilon - \epsilon_\theta(z_t, c, t)\|_2^2]. \quad (3)$$

为了利用丰富的生成先验, HazeGen和DiffDehaze均构建在预训练的Stable Diffusion模型 [39]之上。与在RGB像素空间中运行的传统扩散方法不同, Stable Diffusion在由独立自编码器创建的低维潜在空间中执行扩散和去噪过程。

3.2. 真实有雾图像生成框架

尽管配备了强大的生成先验, HazeGen仍必须有效地整合来自干净图像的条件信息。一个直接的解决方案是在成对的合成图像上训练HazeGen。然而, 我们观察到这种方法会迅速使生成先验退化, 因为使用物理散射模型进行的简单雾合成很容易被识别, 从而被模型学习到。结果, 生成的有雾图像与合成数据高度相似, 导致与真实世界图像之间存在显著的域差距。为了缓解这一问题, 我们提出了专门的混合训练和混合采样算法。

混合训练。为了在逐步增强内容一致性的同时保持生成真实有雾图像的能力, 我们引入了混合训练目标。具体而言, 条件生成目标使用合成图像对, 引导模型在生成的有雾图像与对应的清晰图像之间建立内容关系。相反, 无条件目标使用无标签的真实世界有雾图像, 帮助 HazeGen 保持并进一步增强其生成真实雾气的能力, 从而防止灾难性遗忘。将这两个目标结合起来, 混合训练损失为:

$$\mathcal{L} = p \mathbb{E}_{z^s, c^s, t, \epsilon} [\|\epsilon - \epsilon_\theta(z_t^s, c^s, t)\|_2^2] + (1 - p) \mathbb{E}_{z^r, t, \epsilon} [\|\epsilon - \epsilon_\theta(z_t^r, \emptyset, t)\|_2^2], \quad (4)$$

其中编码后的合成图像对表示为 (z^s, c^s) , 而编码后的真实世界有雾图像则表示为 z^r 。 p 是一个权衡参数, 用于决定应用条件目标的概率。

Algorithm 1 Blended sampling algorithm, given the denoising model ϵ_θ and the VAE’s encoder $\mathcal{E}$ and decoder $\mathcal{D}$

Require: w : mixture coefficient, y : a clean image

```

1:  $\mathbf{z}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
     $\triangleright$  Blended noise prediction
3:  $\hat{\epsilon} = w \epsilon_\theta(\mathbf{z}_t, \mathcal{E}(y), t) + (1 - w) \epsilon_\theta(\mathbf{z}_t, \emptyset, t)$ 
     $\triangleright$  Sampling step
4:  $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
5:  $\mathbf{z}_{t-1} \leftarrow \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{z}_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \hat{\epsilon} \right) + \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t} (1 - \alpha_t) \epsilon$ 
6: end for
7: return  $\mathcal{D}(\mathbf{z}_0)$ 
```

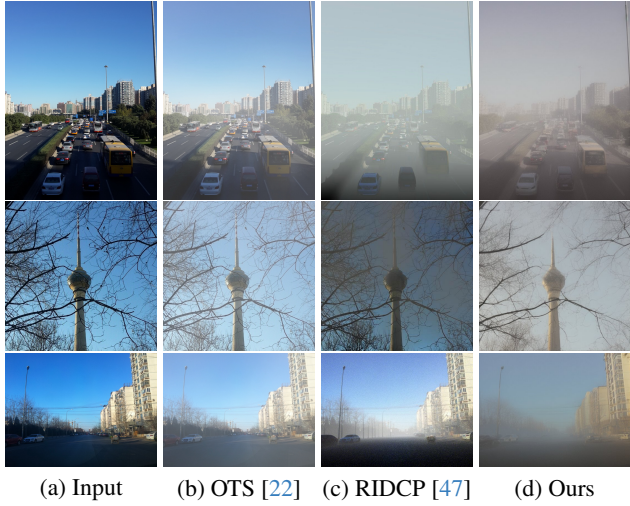


Figure 2. Visual comparisons between hazy images generated by HazeGen and synthetic images from OTS [22] and the phenomenological degradation pipeline of RIDCP [47].

Blended Sampling. Although hybrid training significantly mitigates overfitting to synthetic data, we further propose a blended sampling strategy. This strategy incorporates a small fraction of unconditional predictions into the conditional noise predictions at each sampling step. The advantages of blended sampling are two-fold: (1) unconditional predictions, leveraging intact generative capabilities, can effectively compensate for the learned defects in conditional predictions; and (2) the overall diversity of generated images is notably improved. The detailed blended sampling procedure is provided in Algorithm 1.

An analogy can be made between our approach and the classifier-free diffusion guidance [17]. However, distinct from classifier-free guidance, our primary motivation is to mitigate the adverse impact of low-quality training data and to enhance sampling diversity.

Visual comparisons between hazy images generated by HazeGen and two physical scattering model-based synthe-

sis, OTS [22] and the RIDCP degradation pipeline [47], are presented in Figure 2. The hazy images from OTS exhibit only mild haze, while those from the RIDCP degradation pipeline suffer from unrealistic abrupt haze-density changes around nearby objects and artifacts caused by inaccurate depth estimation. In contrast, HazeGen generates hazy images with more realistic and visually consistent haze.

3.3. Diffusion-based Dehazing Framework

DiffDehaze is trained with high-quality data generated by HazeGen and can thus produce high-quality dehazing results. To reduce the computational cost of lengthy sampling processes and improve fidelity, we propose the Accelerated Fidelity-Preserving Sampling process, AccSamp. As illustrated in Figure 3, the sampling process is divided into two stages: the dehazing estimate generation stage and the guided refinement stage. Specifically, the dehazing estimate generation stage covers initial sampling steps from T down to τ , and the guided refinement stage encompasses the final ω steps. The intermediate timesteps between τ and ω can be skipped to enhance sampling efficiency.

Dehazing Estimate Generation Stage. The core of AccSamp lies in the fast dehazing estimate generation enabled by AlignOp. Inspired by style transfer [19] and image-level normalization [7], we observe that aligning local mean and variance between hazy and corresponding clean image patches yields a reliable preliminary dehazing estimate. This insight aligns well with the physical scattering model: within local regions of approximately uniform scene depth—implying a nearly constant transmission map in Equation 1—the hazy image patch essentially represents a scaled and shifted version of the clean patch. Consequently, statistical alignment approximates a reversal of the haze formation process. Importantly, as only the statistics of clean image patches are required, their details are unnecessary. Therefore, we equivalently utilize an early-stage diffusion prediction, based by the observation that conditional diffusion models establish the overall color distribution within a small fraction of diffusion steps. Through AlignOp, a blurry, low-quality early-stage prediction at timestep τ can be transformed to a satisfactory coarse dehazing estimate with effective dehazing performance while preserving details from the hazy image.

Specifically, the predicted mean at timestep $t = \tau$, denoted as $\hat{z}_0^{(\tau)}$, can be computed by reversing Equation 2:

$$\hat{z}_0^{(\tau)} = \frac{z_\tau - \sqrt{1 - \bar{\alpha}_\tau} \epsilon_\theta(z_\tau, c, \tau)}{\sqrt{\bar{\alpha}_\tau}}, \quad (5)$$

where $c = \mathcal{E}(x)$ represents the encoded hazy image.

Overlapping patches from the hazy image x and the early prediction $\mathcal{D}(\hat{z}_0^{(\tau)})$ are extracted using a sliding window with patch size $k \times k$ and stride d . These patches are denoted

算法1 混合采样算法, 给定去-

噪模型 ϵ_θ 和变分自编码器的编码器 $\mathcal{E}$ 和解码器 $\mathcal{D}$

要求: w : 混合系数, y : 一张干净图像

```

1:  $\mathbf{z}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: 对于  $t = T, \dots, 1$  do
     $\triangleright$  混合噪声预测
3:  $\hat{\epsilon} = w \epsilon_\theta(\mathbf{z}_t, \mathcal{E}(y), t) + (1 - w) \epsilon_\theta(\mathbf{z}_t, \emptyset, t)$ 
     $\triangleright$  采样步骤
4:  $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
5:  $\mathbf{z}_{t-1} \leftarrow \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{z}_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \hat{\epsilon} \right) + \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t} (1 - \alpha_t) \epsilon$ 
6: 结束循环
7: 返回  $\mathcal{D}(\mathbf{z}_0)$ 
```

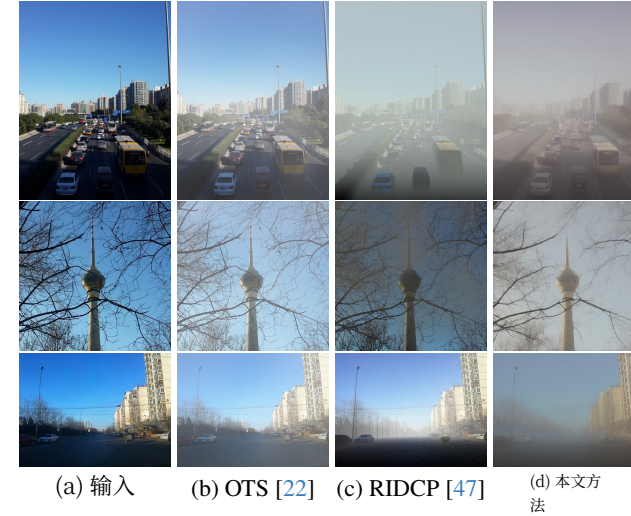


图2. HazeGen生成的有雾图像与OTS [22] 合成图像以及RIDCP [47]现象学退化流程的可视化对比。

混合采样. 尽管混合训练显著减轻了对合成数据的过拟合, 我们进一步提出了一种混合采样策略。该策略在每个采样步骤中将一小部分无条件预测纳入条件噪声预测中。混合采样的优势有两个方面: (1) 无条件预测利用了完好的生成能力, 可以有效补偿条件预测中学习到的缺陷; (2) 生成图像的整体多样性显著提升。详细的混合采样过程在算法1中给出。

我们的方法可以与无分类器扩散引导 [17] 进行类比。然而, 与无分类器引导不同, 我们的主要动机是减轻低质量训练数据的不利影响并增强采样多样性。

HazeGen生成的有雾图像与两种基于物理散射模型的合成方法——

OTS [22]以及RIDCP退化流程 [47], ——的对比如图2所示。OTS生成的有雾图像仅呈现轻度雾霾, 而RIDCP退化流程生成的有雾图像在邻近物体周围存在不真实的雾密度突变, 并因深度估计不准确而产生伪影。相比之下, HazeGen生成的有雾图像具有更真实且视觉一致的雾效。

3.3. 基于扩散的去雾框架

DiffDehaze使用HazeGen生成的高质量数据进行训练, 因此能够产生高质量的去雾结果。为了降低冗长采样过程的计算成本并提高保真度, 我们提出了加速保真采样过程, AccSamp。如图3所示, 采样过程分为两个阶段: 去雾估计生成阶段和引导细化阶段。具体而言, 去雾估计生成阶段涵盖从 T 到 τ 的初始采样步数, 引导细化阶段包含最后 ω 步。介于 τ 和 ω 之间的中间时间步可以被跳过以提高采样效率。

去雾估计生成阶段. Acc-Samp 的核心在于由AlignOp实现的快速去雾估计生成。受风格迁移 [19] 和图像级归一化 [7], 的启发, 我们观察到, 对齐有雾图像与对应干净图像块之间的局部均值和方差, 可以得到可靠的初步去雾估计。这一见解与物理散射模型高度吻合: 在场景深度近似均匀的局部区域内——意味着公式1中的透射率图近乎恒定——有雾图像块本质上可以视为干净图像块的缩放和平移版本。因此, 统计对齐近似于对雾气形成过程的逆操作。重要的是, 由于只需要干净图像块的统计量, 其细节并非必需。因此, 我们等效地利用早期扩散预测, 这是基于条件扩散模型能够在少量扩散步骤内建立整体颜色分布的观察。通过AlignOp, 在时间步 τ 处的模糊、低质量早期预测可以被转化为令人满意的粗略去雾估计, 在保留有雾图像细节的同时实现有效的去雾性能。

具体来说, 在时间步 $t = \tau$ 处预测的均值, 记作 $\hat{z}_0^{(\tau)}$, 可以通过对公式2进行逆运算得到:

$$\hat{z}_0^{(\tau)} = \frac{z_\tau - \sqrt{1 - \bar{\alpha}_\tau} \epsilon_\theta(z_\tau, c, \tau)}{\sqrt{\bar{\alpha}_\tau}}, \quad (5)$$

其中 $c = \mathcal{E}(x)$ 表示编码有雾图像。

使用滑动窗口从有雾图像 x 和早期预测 $\mathcal{D}(\hat{z}_0^{(\tau)})$ 中提取重叠图像块, 其中块大小为 $k \times k$, 步长为 d 。这些图像块记为

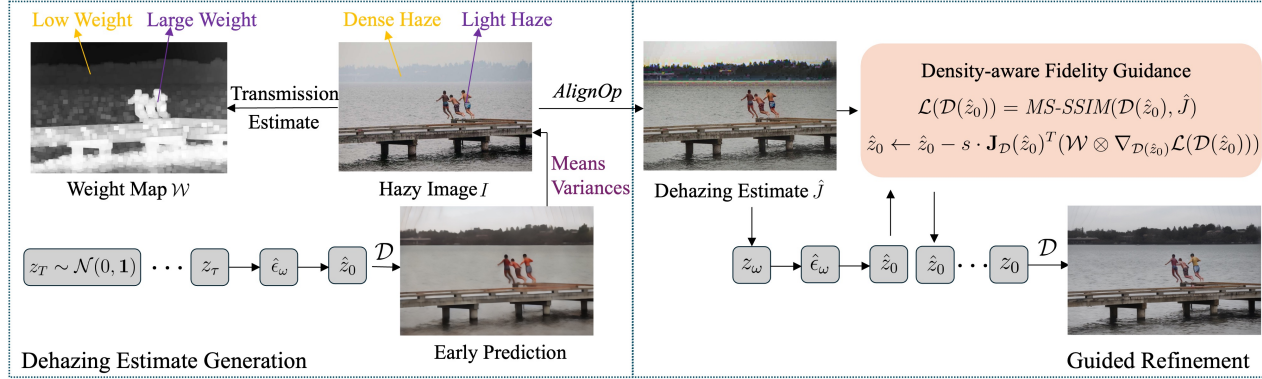


Figure 3. Overview of the AccSamp sampling process. The accelerated sampling process consists of two stages: the dehazing estimate generation stage and the guided refinement stage. In the initial stage (timesteps T to τ), AlignOp transforms a blurry early diffusion prediction into a detailed and faithful dehazing estimate. In the subsequent refinement stage (the final ω steps), additional vivid details are generated under density-aware fidelity guidance. Intermediate sampling steps between τ and ω are skipped to enhance efficiency.

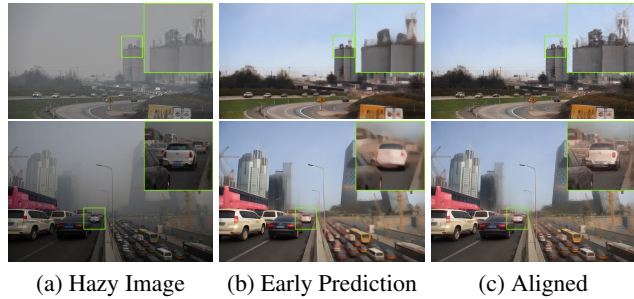


Figure 4. Visualization of AlignOp's effect. By aligning the local patch statistics of the hazy image with those of an early diffusion prediction, AlignOp produces a clean and faithful dehazing estimate.

as $\{p_i^x\}$ and $\{p_i^r\}$, respectively. For each pair of patches (p_i^x, p_i^r) , the aligned patch $p_i^{\hat{y}}$ is computed as

$$p_i^{\hat{y}} = \frac{p_i^x - \mu(p_i^x)}{\sigma(p_i^x)} \cdot \sigma(p_i^r) + \mu(p_i^r), \quad (6)$$

where $\mu(\cdot)$ and $\sigma(\cdot)$ compute the channel-wise mean and standard deviation. The dehazing estimate $\hat{y}$ is then obtained by assembling these aligned patches and averaging their overlapping regions.

Since the dehazing estimate retains details from the hazy image, its difference from the actual underlying clean image is relatively minor. Consequently, we assume that after adding Gaussian noise at timestep ω , the distribution of the dehazing estimate closely approximates that of the true clean image. Thus, z_ω can be directly approximated by adding noise to $\mathcal{E}(\hat{y})$ with $\epsilon \sim \mathcal{N}(0, 1)$:

$$z_\omega = \sqrt{\alpha_\omega} \mathcal{E}(\hat{y}) + \sqrt{1 - \alpha_\omega} \epsilon. \quad (7)$$

At timestep ω , the sampling process resumes from z_ω , refining the result and adding further details, especially in

densely hazy regions. Compared with initiating sampling from purely random noise at timestep $t = T$, our method provides enhanced sampling fidelity because the dehazing estimate $\hat{y}$ derives content from the input image.

Guided Refinement Stage. To further improve sampling fidelity, we propose a haze density-aware fidelity guidance mechanism, which guides the denoising process towards the dehazing estimate $\hat{y}$ during refinement. In general, densely hazy regions benefit more from generated contents, while slightly hazy regions should emphasize fidelity to the input image. To achieve this adaptive weighting, we compute a rough transmission map $\mathcal{W}$ for the hazy input image x using the Dark Channel Prior (DCP) algorithm [16]. This transmission map effectively represents the inverse of haze density because regions with lighter haze exhibit higher transmission values. Consequently, we use $\mathcal{W}$ directly as a weighting map.

At each timestep t , we obtain the predicted clean image $\mathcal{D}(\hat{z}_0^{(t)})$ based on Equation 5. We then employ the MS-SSIM loss [45], which is sensitive to structural similarity, as our fidelity metric. Formally, the fidelity loss is defined as:

$$\mathcal{L}(\mathcal{D}(\hat{z}_0^{(t)})) = MS-SSIM(\mathcal{D}(\hat{z}_0^{(t)}), \hat{y}). \quad (8)$$

We apply gradient descent to optimize $\hat{z}_0^{(t)}$ towards higher fidelity. Specifically, the gradient of the fidelity loss with respect to $\mathcal{D}(\hat{z}_0^{(t)})$ is multiplied elementwise with the weighting map $\mathcal{W}$ to selectively reduce guidance strength in densely hazy regions. At each refinement step, $\hat{z}_0^{(t)}$ is updated according to:

$$\hat{z}_0^{(t)} \leftarrow \hat{z}_0^{(t)} - s \cdot J_{\mathcal{D}}^T(\hat{z}_0^{(t)}) (\mathcal{W} \otimes \nabla_{\mathcal{D}(\hat{z}_0^{(t)})} \mathcal{L}(\mathcal{D}(\hat{z}_0^{(t)}))), \quad (9)$$

where s controls the guidance strength, $J_{\mathcal{D}}(\hat{z}_0^{(t)})$ is the Jacobian matrix of the decoder $\mathcal{D}$, and $\otimes$ denotes elementwise

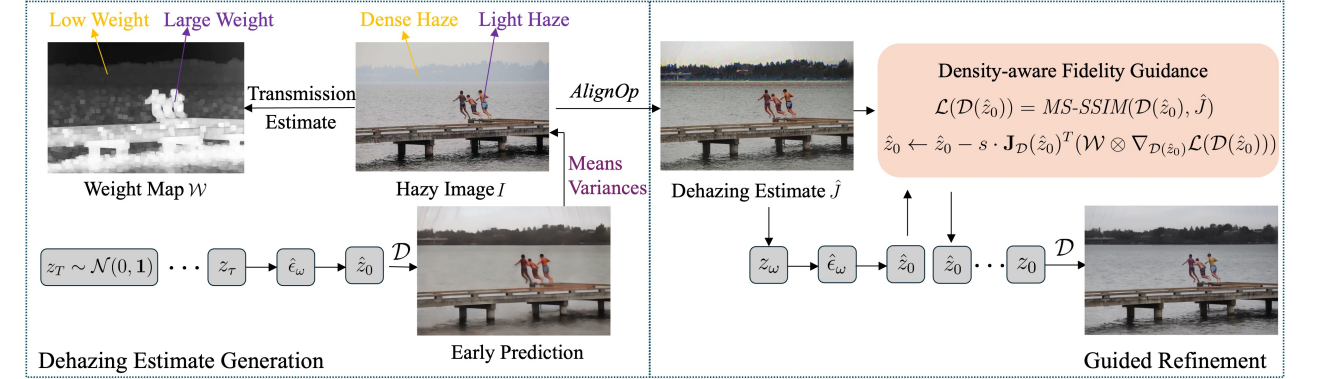


图3. AccSamp采样过程概览。加速采样过程包含两个阶段：去雾估计生成阶段和引导细化阶段。在初始阶段（时间步 T 至 τ ），AlignOp将模糊的早期扩散预测转换为细节丰富且保真的去雾估计。在后续细化阶段（最后 ω 步），在密度感知保真引导下生成额外的生动细节。介于 τ 和 ω 之间的中间采样步骤被跳过以提升效率。

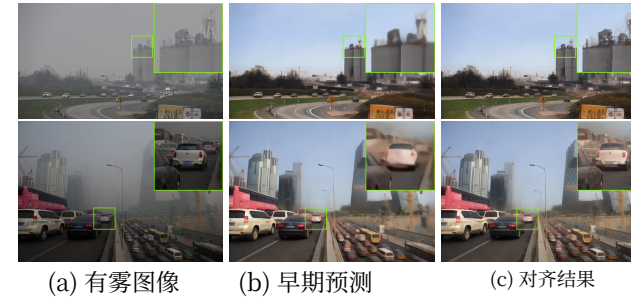


图4. AlignOp效果可视化。通过将有雾图像的局部图像块统计与早期扩散预测的相应统计对齐，AlignOp生成了干净且保真的去雾估计。

$\{p_i^x\}$ 和 $\{p_i^r\}$ 。对于每对图像块 (p_i^x, p_i^r) ，对齐结果 $p_i^{\hat{y}}$ 的计算方式为

$$p_i^{\hat{y}} = \frac{p_i^x - \mu(p_i^x)}{\sigma(p_i^x)} \cdot \sigma(p_i^r) + \mu(p_i^r), \quad (6)$$

其中 $\mu(\cdot)$ 和 $\sigma(\cdot)$ 分别计算逐通道均值和标准差。随后通过组装这些对齐后的图像块并对其重叠区域取均值，得到去雾估计 $\hat{y}$ 。

由于去雾估计保留了有雾图像中的细节，其与实际底层干净图像之间的差异相对较小。因此，我们假设在时间步 ω 处添加高斯噪声后，去雾估计的分布会非常接近真实干净图像的分布。因此， z_ω 可以直接通过向 $\mathcal{E}(\hat{y})$ 添加带有 $\{5\}(0, 1)$ 的噪声来近似：

$$z_\omega = \sqrt{\alpha_\omega} \mathcal{E}(\hat{y}) + \sqrt{1 - \alpha_\omega} \epsilon. \quad (7)$$

在时间步 ω ，采样过程从 z_ω 恢复，细化结果并添加更多细节，尤其是在浓雾区域。

与在时间步 $t = T$ 从纯随机噪声开始采样相比，我们的方法提供了更高的采样保真度，因为去雾估计 $\hat{y}$ 的内容来源于输入图像。

引导细化阶段。 为了进一步提高采样保真度，我们提出了一种雾密度感知保真引导机制，该机制在细化过程中引导去噪过程趋向于去雾估计 $\hat{y}$ 。一般来说，浓雾区域会更多受益于生成内容，而轻度有雾区域则应侧重于对输入图像的保真度。为了实现这种自适应加权，我们使用暗通道先验 (DCP) 算法 [16] 为有雾输入图像 x 计算粗略的透射率图 $\mathcal{W}$ 。该透射率图有效地表示了雾密度的倒数，因为雾较轻的区域具有较高的透射率值。因此，我们直接使用 $\mathcal{W}$ 作为权重图。

在每个时间步 t ，我们基于公式 5 获得预测的清晰图像 $\mathcal{D}(\hat{z}_0^{(t)})$ 。随后，我们采用对结构相似性敏感的 MS-SSIM 损失 [45]，作为保真度量。形式上，保真损失定义为：

$$\mathcal{L}(\mathcal{D}(\hat{z}_0^{(t)})) = MS-SSIM(\mathcal{D}(\hat{z}_0^{(t)}), \hat{y}). \quad (8)$$

我们应用梯度下降来优化 $\hat{z}_0^{(t)}$ ，使其具有更高的保真度。具体来说，保真损失关于 $\mathcal{D}(\hat{z}_0^{(t)})$ 的梯度与权重图 $\mathcal{W}$ 逐元素相乘，以在浓雾区域选择性地降低引导强度。在每个细化步骤中， $\hat{z}_0^{(t)}$ 根据以下公式更新：

$$\hat{z}_0^{(t)} \leftarrow \hat{z}_0^{(t)} - s \cdot J_{\mathcal{D}}^T(\hat{z}_0^{(t)}) (\mathcal{W} \otimes \nabla_{\mathcal{D}(\hat{z}_0^{(t)})} \mathcal{L}(\mathcal{D}(\hat{z}_0^{(t)}))), \quad (9)$$

其中 s 控制引导强度， $J_{\mathcal{D}}(\hat{z}_0^{(t)})$ 是解码器 $\mathcal{D}$ 的雅可比矩阵，而 $\otimes$ 表示逐元素

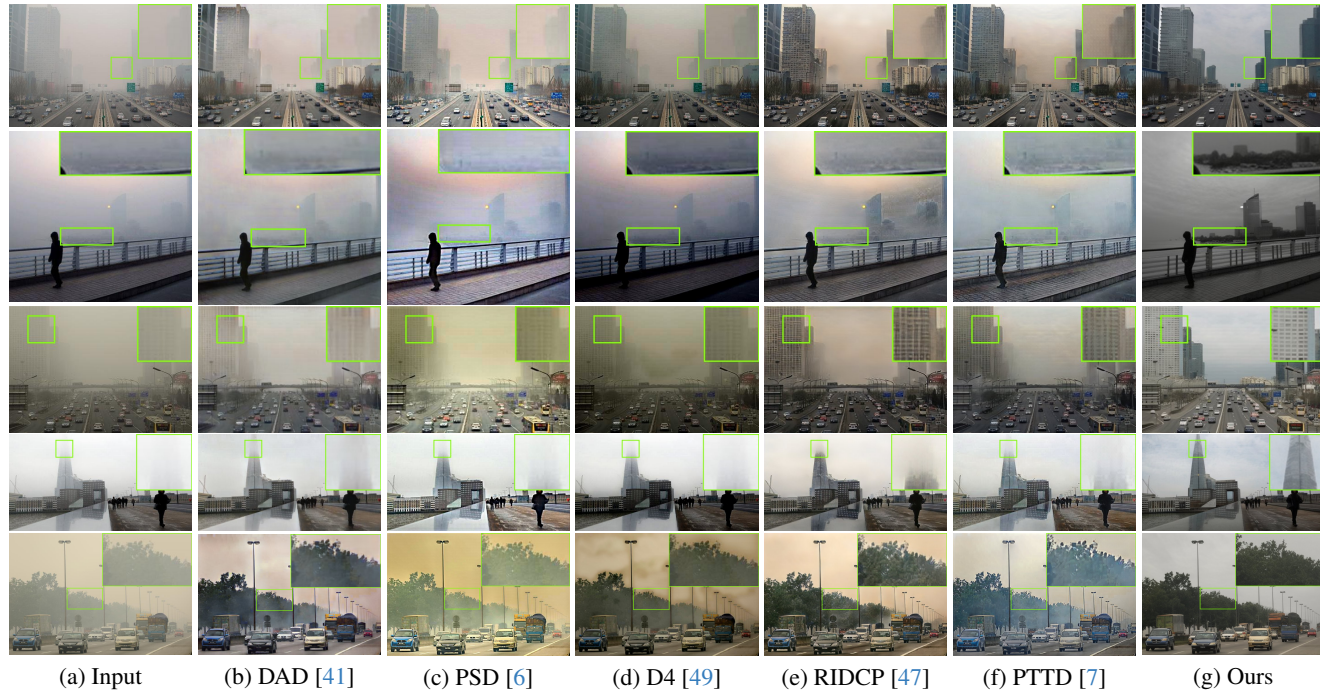


Figure 5. Visual comparisons on the RTTS dataset [22]. Zoomed-in for details.

multiplication. In this scheme, the guidance strength s offers a trade-off between generated image *quality* and *fidelity* to the hazy image. Finally, z_{t-1} is computed using the updated $\hat{z}_0^{(t)}$ and z_t , after which the sampling process moves to the next timestep.

4. Experiments

4.1. Experiment Settings

Datasets. Around 4,800 real-world hazy images from the URHI split of the RESIDE dataset [22], together with synthetic image pairs generated by the phenomenological degradation pipeline from RIDCP [47], were used for the training of HazeGen. Subsequently, we generated 100,000 realistic hazy images with HazeGen to provide high-quality training data for DiffDehaze. For qualitative and quantitative evaluations, we employed the widely used RTTS split from RESIDE [22], which contains 4,322 images covering diverse scenes and haze patterns. The Fattal’s dataset [13] is used for additional visual comparison, which comprises 31 classical hazy images.

Implementation Details. The proposed pipeline is implemented using PyTorch 2.2.2 and is built upon Stable Diffusion v2-1 [39]. HazeGen is optimized with the AdamW optimizer [29], employing a learning rate of 3×10^{-5} . The probability parameter p is set to 0.3, the training batch size is 16, and training lasts for 5,000 iterations. For the sampling of HazeGen, we adopt the spaced DDPM sampler [33]

with 50 steps and set the mixture coefficient w to 0.85. The training of DiffDehaze adopts the same optimizer and learning rate but lasts for 55,000 iterations. To keep optimal image quality, DiffDehaze employs AccSamp with 50 steps and empirically set parameters $\tau = 800$, $\omega = 600$, and guidance strength $s = 0.1$. It’s worth noting that reducing ω or increasing s still produces satisfactory results.

4.2. Comparison with State-of-the-Art Methods

We evaluate the proposed method against several state-of-the-art approaches for real-world image dehazing, including DAD [41], PSD [6], D4 [49], RIDCP [47], and PTTD [7]. Comprehensive experiments are conducted to provide a thorough assessment.

Quantitative Comparisons. As real-world hazy datasets lack ground-truth clean images, we adopt multiple no-reference image quality metrics for quantitative evaluation. A detailed evaluation on the RTTS dataset using FADE [8], Q-Align [46], LIQE [51], CLIPQA [43], ManIQA [48], MUSIQ [21], and BRISQUE [31] is presented in Table 1. Our method achieves leading performance across all metrics except for FADE. Specifically, it attains a remarkable 23.8% improvement on the Q-Align metric, as well as substantial improvements on other recent metrics such as LIQE and CLIPQA. However, our method has relatively worse performance on FADE, which is partly due to FADE’s unreliability in evaluating dehazing quality. The limitations of FADE will be further discussed in the next section. Over-

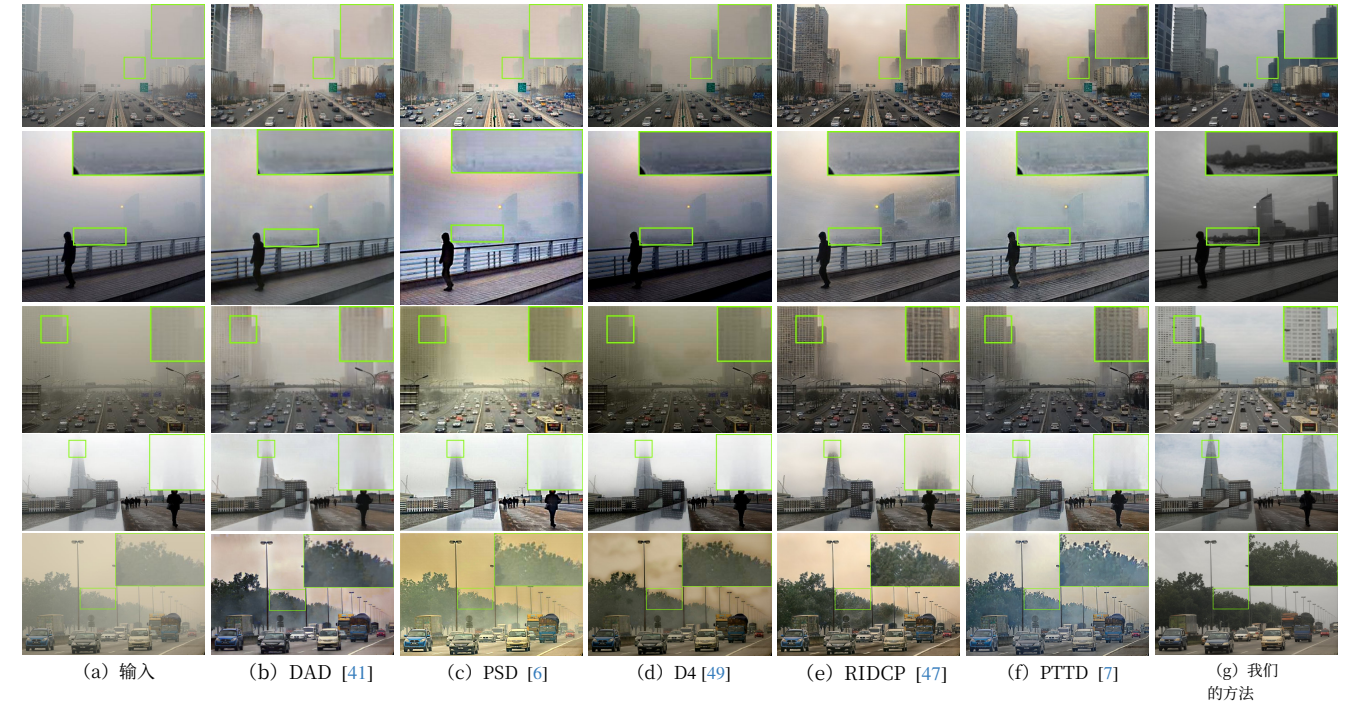


图5. 在RTTS数据集 [22]上的视觉对比。放大以查看细节。

乘法。在该方案中，引导强度 s 在生成图像质量与对有雾图像的保真度之间提供了一种权衡。最后，使用更新后的 $\hat{z}_0^{(t)}$ 和 z_t 计算 z_{t-1} ，随后采样过程进入下一个时间步。

4. 实验

4.1. 实验设置

数据集。 我们使用来自RESIDE数据集 [22], URHI划分的约4800张真实世界有雾图像，以及由RIDCP [47], 的现象学退化流程生成的合成图像对来训练 HazeGen。随后，我们使用HazeGen生成了100000张真实感有雾图像，为DiffDehaze提供高质量的训练数据。对于定性和定量评估，我们采用了RESIDE [22], 中广泛使用的RTTS划分，该划分包含4322张图像，覆盖了多样的场景和雾型。Fattal 数据集 [13]用于额外的视觉对比，其中包含31张经典的有雾图像。

实现细节。 所提出的流程使用 PyTorch 2.2.2 实现，并基于 Stable Diffusion v2-1 [39] 构建。HazeGen 使用 AdamW 优化器 [29], 进行优化，采用的学习率为 3×10^{-5} 。概率参数 p 设为 0.3，训练批量大小为 16，训练共进行 5,000 次迭代。对于 HazeGen 的采样，我们采用间隔DDPM 采样器 [33]

共 50 步，并将混合系数 w 设为 0.85。DiffDehaze 的训练采用相同的优化器和学习率，但共进行 55,000 次迭代。为保持最佳图像质量，DiffDehaze 使用 AccSamp 并设置 50 步，并根据经验设置参数 $\tau = 800$, $\omega = 600$ 以及引导强度 $s = 0.1$ 。值得注意的是，降低 ω 或提高 s 仍能产生令人满意的结果。

4.2. 与最先进方法的对比

我们将所提出的方法与几种最先进的真实世界图像去雾方法进行了评估对比，包括DAD [41], PSD [6], D4 [49], RIDCP [47], 和PTTD [7]。通过全面的实验来提供详尽的评估。

定量比较。 由于真实世界有雾数据集缺乏真实清晰图像，我们采用多种无参考图像质量指标进行定量评估。在RTTS数据集上使用FADE [8], Q-Align [46], LIQE [51], CLIPQA [43], ManIQA [48], MUSIQ [21], 和 BRISQUE [31] 的详细评估结果见表1。除FADE外，我们的方法在所有指标上均取得了领先性能。具体而言，它在Q-Align指标上取得了显著的23.8%提升，同时在LIQE和CLIPQA等近期提出的指标上也有大幅改进。然而，我们的方法在FADE上的表现相对较差，部分原因是FADE在评估去雾质量方面不够可靠。FADE的局限性将在下一节中进一步讨论。总

Table 1. Quantitative comparisons of various dehazing methods on the RTTS dataset [22]. **Bold** numbers indicate the best performance.

Method	Venue	FADE↓	Q-Align↑	LIQE↑	CLIPQA↑	ManIQA↑	MUSIQ ↑	BRISQUE↓
Hazy Input	-	2.484	2.0586	1.9146	0.3882	0.3081	53.768	36.6423
DAD [41]	CVPR’20	1.130	2.0117	1.6666	0.2512	0.2219	49.337	32.4565
PSD [6]	CVPR’21	0.920	1.9014	1.5174	0.2497	0.2640	52.806	21.6160
D4 [49]	CVPR’22	1.358	2.0801	1.9741	0.3404	0.2974	53.555	28.1015
RIDCP [47]	CVPR’23	0.944	2.4844	2.5518	0.3367	0.2769	59.384	17.2944
PTTD [7]	ECCV’24	0.712	2.2891	2.3532	0.3700	0.3095	62.114	16.6302
DiffDehaze (Ours)	-	1.138	2.8340	3.0693	0.4263	0.3661	65.086	16.4924

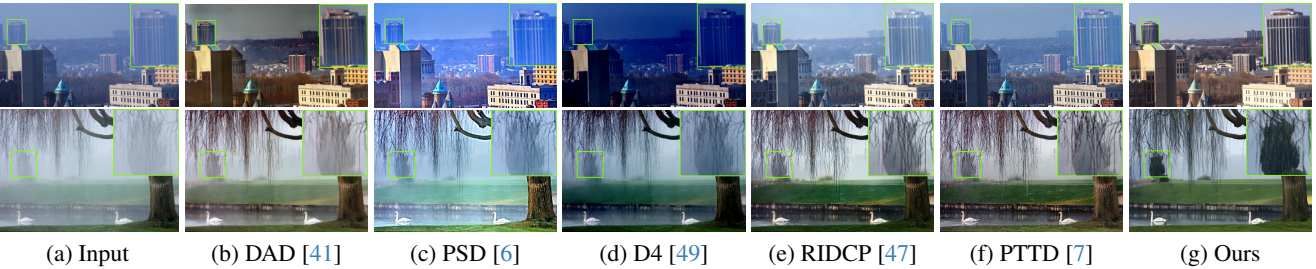


Figure 6. Visual comparisons on the Fattal’s dataset [13]. Zoomed-in for details.

Table 2. Ablation study of HazeGen on the RTTS dataset [22]. **Bold** numbers indicate the best performance.

Variant	Q-Align↑	LIQE↑	CLIPQA↑	ManIQA↑	MUSIQ ↑
w/o hybrid	2.6582	2.7754	0.4174	0.3513	63.079
w/o blended	2.5410	2.6410	0.4262	0.3510	62.489
w/o both	2.8223	1.5983	0.3266	0.2782	49.933
w/o HazeGen	2.1660	2.0014	0.3540	0.2928	55.871
full version	2.8340	3.0693	0.4263	0.3661	65.086

all, our approach demonstrates superior quantitative performance compared to existing methods.

Qualitative Comparisons. Qualitative results comparing our method and state-of-the-art real-world dehazing methods on the RTTS dataset [22] are illustrated in Figure 5. Notably, our method is capable of recovering vivid details (e.g., the trees in the fifth row), whereas other methods fail to produce such details. Additional qualitative comparisons using Fattal’s dataset [13] are shown in Figure 6. Images generated by DAD, PSD, D4, RIDCP, and PTTD often contain residual haze, especially in heavily hazy areas. In contrast, our method consistently removes haze from all regions, producing visually appealing images with natural color restoration that closely resemble clear weather conditions.

4.3. Ablation Studies

Ablation Study for HazeGen. To verify the effectiveness of the proposed training and sampling strategies of HazeGen, we evaluate the performance of DiffDehaze trained on data generated by several variants of HazeGen: (a) with-

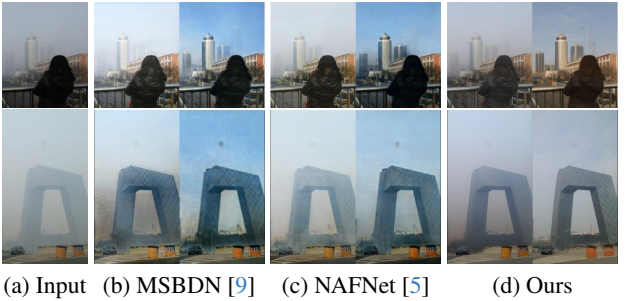


Figure 7. Comparison of training data effectiveness. The left half of each image shows results from models trained on synthetic hazy images from RIDCP [47], while the right half shows results from models trained on realistic hazy images generated by HazeGen.

out hybrid training (i.e., purely conditional training); (b) without blended sampling; (c) without both hybrid training and blended sampling; and (d) without HazeGen, meaning DiffDehaze is trained directly on synthetic data. The results on the RTTS dataset, presented in Table 2, demonstrate that each component is essential to achieve optimal performance. Visual comparisons are provided in the supplementary material.

Effectiveness of HazeGen. To further validate the quality of images generated by HazeGen, we compare the dehazing performance of two popular models—MSBDN [9] and NAFNet [5]—as well as DiffDehaze, trained separately with synthetic data and data generated by HazeGen. As illustrated in Figure 7, all models trained with data generated

表1. 定量比较

各种去雾方法在RTTS数据集上的比较 [22]。粗体数字表示最佳性能。

方法	发表来源	FADE↓	Q-Align↑	LIQE↑	CLIPQA↑	ManIQA↑	MUSIQ ↑	BRISQUE↓
有雾输入	-	2.484	2.0586	1.9146	0.3882	0.3081	53.768	36.6423
DAD [41]	CVPR’ 20	1.130	2.0117	1.6666	0.2512	0.2219	49.337	32.4565
PSD [6]	CVPR’ 21	0.920	1.9014	1.5174	0.2497	0.2640	52.806	21.6160
D4 [49]	CVPR’ 22	1.358	2.0801	1.9741	0.3404	0.2974	53.555	28.1015
RIDCP [47]	CVPR’ 23	0.944	2.4844	2.5518	0.3367	0.2769	59.384	17.2944
PTTD [7]	ECCV’ 24	0.712	2.2891	2.3532	0.3700	0.3095	62.114	16.6302
DiffDehaze (本文方法)	-	1.138	2.8340	3.0693	0.4263	0.3661	65.086	16.4924

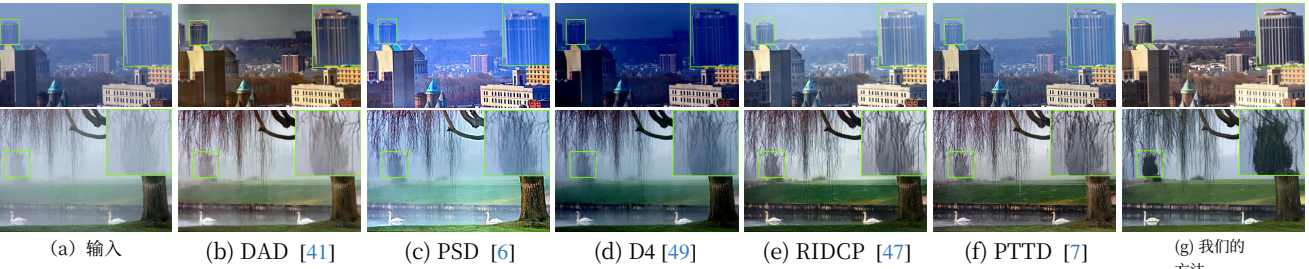


图6. 在Fattal数据集 [13]上的视觉对比。放大查看细节。

表2. HazeGen在RTTS数据集 [22]上的消融研究。粗体数字表示最佳性能。

变体	Q-Align↑	LIQE↑	CLIPQA↑	ManIQA↑	MUSIQ ↑
无混合	2.6582	2.7754	0.4174	0.3513	63.079
无混合	2.5410	2.6410	0.4262	0.3510	62.489
两者皆无	2.8223	1.5983	0.3266	0.2782	49.933
无HazeGen	2.1660	2.0014	0.3540	0.2928	55.871
完整版本	2.8340	3.0693	0.4263	0.3661	65.086

体而言，与现有方法相比，我们的方法展现出了优越的定量性能。

定性比较。将我们的方法与最先进的真实世界去雾方法在RTTS数据集上的定性结果 [22] 如图5所示。值得注意的是，我们的方法能够恢复生动细节（例如第五行中的树木），而其他方法则无法产生此类细节。使用Fattal数据集的额外定性比较 [13] 如图6所示。DAD、PSD、D4、RIDCP和PTTD生成的图像通常包含残余雾霾，尤其是在重度雾霾区域。相比之下，我们的方法能够一致地去除所有区域的雾霾，生成视觉上令人愉悦且色彩还原自然的图像，与晴朗天气条件高度相似。

4.3. 消融研究

HazeGen消融研究。为验证所提出的HazeGen训练与采样策略的有效性，我们评估了DiffDehaze在HazeGen若干变体生成数据上训练后的性能：(a) 无

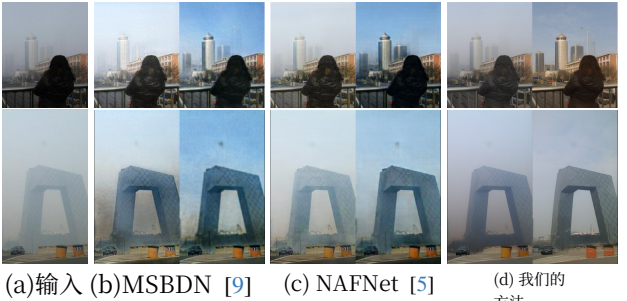


图7. 训练数据有效性对比。每张图像的左半部分展示了在RIDCP [47]，生成的合成有雾图像上训练的模型结果，而右半部分展示了在HazeGen生成的真实感有雾图像上训练的模型结果。

混合训练（即纯条件训练）；(b) 无混合采样；(c) 同时无混合训练和混合采样；(d) 无HazeGen，即DiffDehaze直接在合成数据上训练。如表2所示，RTTS数据集上的结果表明，每个组件对于实现最优性能都至关重要。视觉对比见补充材料。

HazeGen 的有效性。为了进一步验证 HazeGen 生成图像的质量，我们比较了两种主流模型——MSBDN [9]和 NAFNet [5]——以及 DiffDehaze，分别使用合成数据和 HazeGen 生成的数据进行训练后的去雾性能。如图7所示，所有使用 HazeGen 生成数据训练的模型

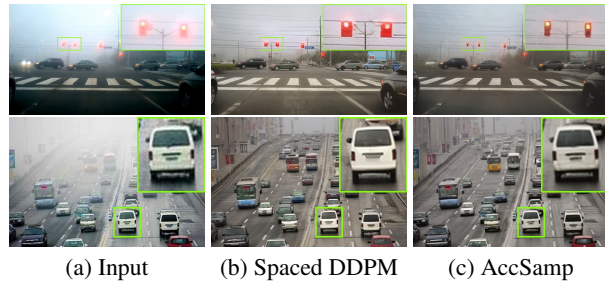


Figure 8. Comparison of sampling results with spaced DDPM [33] and AccSamp samplers.

Table 3. Ablation study of DiffDehaze on the RTTS dataset [22]. **Bold** numbers indicate the best performance.

Variant	FADE↓	Q-Align↑	LIQE↑	CLIPQA↑	ManIQA↑
w/o AlignOp	1.450	2.5645	2.6786	0.4169	0.3617
w/o weighting	1.236	2.8067	2.9769	0.4221	0.3457
w/o both	1.760	2.5079	2.5837	0.4123	0.3337
full version	1.138	2.8340	3.0693	0.4263	0.3661

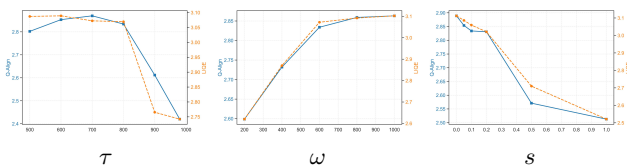


Figure 9. Influence of hyperparameters on dehazing performance measured by Q-Align and LIQE.

by HazeGen exhibit improved dehazing capability and overall visual quality.

Ablation Study for DiffDehaze. Table 3 reports the ablation results for AccSamp, specifically assessing performance without AlignOp or the haze density weighting mechanism. These results highlight the importance of each individual component. In the absence of AlignOp, the fidelity guidance directly uses the hazy input image for loss computation.

Effectiveness of Fidelity Enhancement. Both AlignOp and the adaptive fidelity guidance mechanism enhance the sampling fidelity of AccSamp. To illustrate this improvement, Figure 8 compares sampling results obtained with the standard spaced DDPM sampler [33] against AccSamp. The results clearly show that AccSamp significantly increases fidelity to the input images while preserving effective dehazing capability.

Influence of Hyperparameters in AccSamp. Figure 9 presents quantitative metric comparisons (Q-Align and LIQE) across varying hyperparameter settings in AccSamp. Timesteps τ and ω are optimized to enhance sampling efficiency without compromising performance, while the guidance strength s balances the trade-off between *quality* and

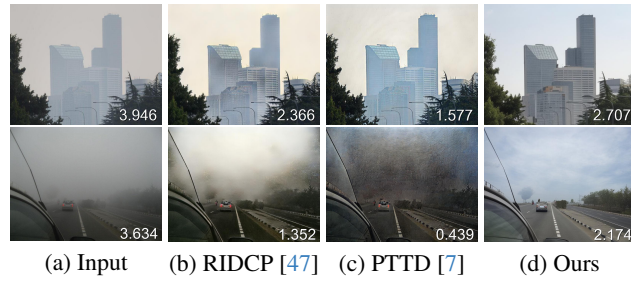


Figure 10. Typical failure cases of FADE evaluation [8]. FADE scores are shown at the bottom-right corner of each image.

fidelity. Therefore, the default hyperparameter configuration is chosen as $\tau = 800$, $\omega = 600$, and $s = 0.1$.

Dehazing Evaluation with FADE [8]. Although our method achieves superior dehazing capability, it shows comparatively lower performance in terms of FADE. This discrepancy primarily arises because FADE is insensitive to color-distorted haze residuals. An experiment comparing FADE evaluation results between our method and two state-of-the-art methods is illustrated in Figure 10. In the first row, RIDCP and PTTD merely shift the haze color toward blue without effective haze removal. In the second row, their results appear visually messy and unclear. However, FADE incorrectly scores these flawed outputs better than ours, contradicting human visual perception.

5. Conclusion

In this work, we introduce HazeGen, a novel framework for realistic hazy image generation, and DiffDehaze, a diffusion-based dehazing framework, building upon controlled Stable Diffusion [39] models. By effectively exploiting generative diffusion priors of natural hazy images and employing hybrid training and blended sampling strategies, HazeGen is capable of generating realistic and high-quality hazy images for the training of DiffDehaze. Moreover, leveraging fast dehazing estimates provided by AlignOp, AccSamp can reduce sampling steps and enhance fidelity for DiffDehaze. Comprehensive experiments demonstrated the superior performance of our approach.

Limitations. Although the proposed method achieves remarkable performance, we identify two notable limitations. First, there remains an urgent need for the development of more sophisticated and reliable metrics for the evaluation of dehazing results. Second, while AccSamp effectively improves sampling fidelity, there is still scope for further enhancement in future research.

Acknowledgment

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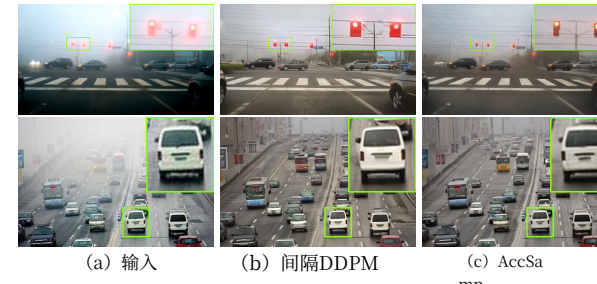


图8. 间隔DDPM [33] 与AccSamp采样器的采样结果对比。

表3. DiffDehaze在RTTS数据集 [22]上的消融研究。加粗数字表示最佳性能。

变体	FADE↓	Q-Align↑	LIQE↑	CLIPQA↑	ManIQA↑
无 AlignOp	1.450	2.5645	2.6786	0.4169	0.3617
无加权	1.236	2.8067	2.9769	0.4221	0.3457
两者皆无	1.760	2.5079	2.5837	0.4123	0.3337
完整版本	1.138	2.8340	3.0693	0.4263	0.3661

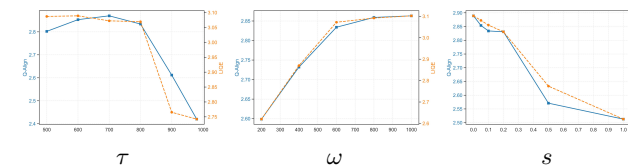


图9. 超参数对去雾性能的影响（通过Q-Align和LIQE衡量）。

均表现出更好的去雾能力和整体视觉质量。

DiffDehaze的消融研究。表3 报告了AccSamp的消融结果，具体评估了在去掉AlignOp或雾霾密度加权机制时的性能。这些结果凸显了每个独立组件的重要性。在缺少AlignOp的情况下，保真度引导直接使用有雾输入图像进行损失计算。

保真度增强的有效性。AlignOp和自适应保真度引导机制都增强了AccSamp的采样保真度。为了说明这种改进，图8 将标准间隔DDPM采样器 [33] 的结果与AccSamp进行了比较。结果清楚地表明，AccSamp显著提高了对输入图像的保真度，同时保持了有效的去雾能力。

AccSamp 中超参数的影响。图9展示了 AccSamp 在不同超参数设置下的定量指标对比（Q-Align 和 LIQE）。时间步 τ 和 ω 经过优化，以在不牺牲性能的前提下提高采样效率，而引导强度 s 则在质量与

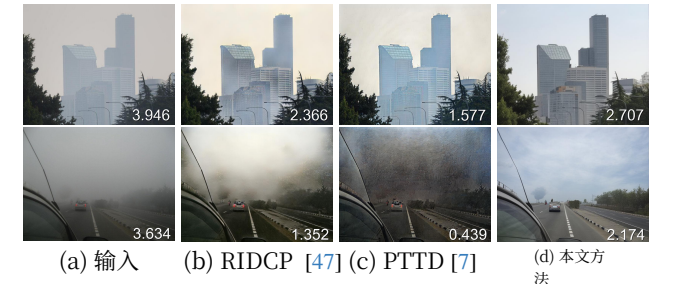


图10. FADE评估的典型失败案例 [8]。FADE分数显示在每张图像的右下角。

保真度之间进行权衡。因此，默认的超参数配置选择为 $\tau = 800$ 、 $\omega = 600$ 和 $s = 0.1$ 。

使用FADE进行去雾评估 [8]。尽管我们的方法实现了卓越的去雾能力，但在FADE指标上表现相对较低。这种差异主要源于FADE对色彩失真的雾霾残差不敏感。图10展示了我们的方法与两种最先进方法之间的FADE评估结果对比实验。在第一行中，RIDCP和PTTD仅将雾霾颜色向蓝色偏移，并未有效去除雾霾。在第二行中，它们的结果在视觉上显得杂乱且不清晰。然而，FADE错误地将这些有缺陷的输出评为优于我们的结果，这与人类视觉感知相矛盾。

5. 结论

在这项工作中，我们引入了HazeGen——一种用于真实感有雾图像生成的新框架，以及DiffDehaze——一种基于扩散的去雾框架，两者均构建在受控 StableDiffusion [39] 模型之上。通过有效利用自然有雾图像的生成扩散先验，并采用混合训练和混合采样策略，HazeGen能够生成真实且高质量的有雾图像，用于 DiffDehaze 的训练。此外，利用 AlignOp 提供的快速去雾估计，AccSamp 可以减少采样步数并增强 DiffDehaze 的保真度。全面的实验证明了我们方法的优越性能。

局限性。尽管所提出的方法取得了显著性能，但我们发现两个值得注意的局限。首先，开发更精密、更可靠的去雾结果评价指标仍然是一项迫切需求。其次，虽然AccSamp有效提升了采样保真度，但在未来研究中仍有进一步改进的空间。

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